

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
4	(a)		Increase in temperature - kinetic theory effects Molecules move randomly Collisions become more frequent [when heat supplied] [No change in volume so heat flowing in] causes increase in U / kinetic energy No work done [as constant volume] T increases with U as temperature proportional to U, or equivalent Newton's laws of motion Momentum of molecules increase Force on molecules = rate of change of momentum (during collision with wall) Force on wall is equal and opposite to force on molecules Greater forces during the collisions Increase in pressure Molecules collide with walls exerting force on walls and / or pressure Pressure increases with temperature Pressure = force on walls per unit area Mean pressure due to many collisions [and many molecules] 5-6 marks Comprehensive account including reference to increase in temperature, Newton's laws of motion and increase in pressure. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured.</i> 3-4 marks Comprehensive account including reference to 2 out of 3 of increase in temperature, Newton's laws of motion and increase in pressure or brief account of all 3 areas. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure.</i>	6			6		

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			1-2 marks Comprehensive account including reference to one of increase in temperature, Newton's laws of motion and increase in pressure or limited account of 2 areas. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)	$\text{rms} = \sqrt{\frac{400^2 + 425^2 + 450^2 + 550^2 + 625^2}{5}} \quad (1)$ $= 497 \text{ [m s}^{-1}\text{]} \quad (1)$		2		2	2	
		(ii)	The expected rms is explained by: $PV = nRT \quad \text{so} \quad P = \frac{nRT}{V}$ $\rho = \text{mass}/V = \frac{n(M_r \times 10^{-3})}{V} \quad \text{use of both equations by substitution (1)}$ substitute these into: $P = \frac{1}{3}\rho \overline{c^2}$; $\sqrt{\overline{c^2}} = \sqrt{\frac{3P}{\rho}}$; $\sqrt{\overline{c^2}} = \sqrt{3 \frac{nRT}{V} \frac{V}{n(M_r \times 10^{-3})}} = \sqrt{\frac{3RT}{(M_r \times 10^{-3})}}$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3(8.31)(293)}{(32 \times 10^{-3})}} = 478 \text{ [m s}^{-1}\text{]} \quad (1)$ Valid conclusion with data i.e. yes, the rms speed of the five molecules is slightly higher [about 4% above the expected rms of the gas] (1) ecf Alternative solution: Use $m\overline{c^2} = 3kT$ $\sqrt{\overline{c^2}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3(1.38 \times 10^{-23})(293)}{(32)(1.66 \times 10^{-27})}}$ with sensible substitution (1) = 478 [m s⁻¹] (1)		1	1	3	3	

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		(iii)	Density does not change and Pressure increases from the original pressure p to $1.2p$ (1) i.e. an increase of 20%. so $\left(\sqrt{c^2}\right)_{\text{new}} = \sqrt{\frac{3(1.2p)}{\rho}}$ $\left(\sqrt{c^2}\right)_{\text{new}} = \sqrt{1.2} \sqrt{\frac{3p}{\rho}} = \sqrt{1.2} (478) = 524 \text{ m s}^{-1} \quad (1)$		2		2	2	
			Question 4 total	6	5	2	13	7	0